

Student Name: Andrew Rodriguez

PI Name: Dr. Nicholas Todd

Primary Mentor Name: Dr. Nicholas Todd (day to day contact Dr. Bernie Owusu-Yaw)

Title of Project: Machine Learning Based Acoustic Feedback Controller for FUS-BBB Mediated AAV Delivery

Project Motivation: The blood-brain barrier (BBB) is a major obstacle for the delivery of novel gene therapies, such as Adeno-Associated Viruses (AAVs), to the central nervous system. Focused Ultrasound Blood-Brain Barrier (FUS-BBB) opening is a promising, non-invasive technique that uses targeted acoustic waves and circulating microbubbles to temporarily increase BBB permeability. However, ensuring both the safety and efficacy of this procedure requires precise control over the ultrasound dose. Currently, Dr. Todd's lab has a need for real time monitoring to prevent under-dosing (ineffective delivery) or over-dosing (tissue damage). As the microbubbles cavitate during FUS, they generate distinct acoustic emissions which can be measured by the piezoelectric crystals of an ultrasound system. The overarching goal of this research is to design a dynamic feedback controller capable of processing these acoustic emissions in real time to predict and control the quantity of AAV delivered to the brain.

Detailed Description of Your Specific Project:

During this 13 week independent study, my project will focus on the first two phases of designing this acoustic feedback controller, heavily utilizing principles of signal processing, machine learning, and quantitative biology.

Specifically, I will process raw, time series acoustic emission data generated from previous FUS experiments (some of this data has already been collected, some more may be collected this summer by Dr. Todd's group). Using Python, I will design feature extraction pipelines, such as the Fourier transform (which shows preliminary promise on the second harmonic), to isolate the specific acoustic signatures correlated with BBB opening. Additionally, I will engage in some hands-on measurement and experimental validation by assisting with in vivo FUS experiments in mouse models. Following the physical experiments, I will perform tissue sectioning, staining, and fluorescence microscopy to quantitatively measure the ground truth concentration of AAVs successfully delivered to the target tissue.

By the end of the term, I will integrate these two workflows by applying machine learning models to map the extracted acoustic features (the input) to the physical AAV delivery outcomes (the output). Through this project, I will gain practical experience in bridging computational modeling with wet-lab biological measurement, applying my coursework in Computer Science and Biomedical Engineering to a translational medicine problem.

Final Deliverable:

A detailed written report, formatted as a draft journal manuscript, detailing the feature extraction methodology, the experimental protocol, and the preliminary predictive accuracy of the baseline

machine learning model. Additionally, dependent upon need in the fall as well as model accuracy, I will make the model easily accessible and deployable to Dr. Todd's group,

Preliminary Weekly Schedule

Tuesday: 1:00 PM - 5:00 PM

Wednesday: 10:00 AM - 12:00 PM

Thursday: 10:00 AM - 2:00 PM

Week 1: Complete lab safety training (BWH/HMS), review relevant literature on FUS-BBB and acoustic cavitation, and finalize computational environment setup.

Week 2: Familiarize with existing raw acoustic emission datasets; begin writing initial Python scripts for data import and basic filtering.

Week 3: Develop and test feature extraction pipelines (e.g., Fourier transforms) on existing acoustic data.

Week 4: Refine feature extraction algorithms to isolate harmonic, subharmonic, and broadband emission signatures.

Week 5: Begin assisting with in vivo FUS experiments; observe and learn the focused ultrasound protocols.

Week 6: Continue assisting with in vivo experiments; begin training on wet lab techniques for post experiment tissue handling.

Week 7: Perform brain sectioning and biological staining on tissue from recent FUS experiments to prepare for AAV quantification.

Week 8: Utilize fluorescence microscopy and imaging software to quantitatively measure physical AAV delivery in stained tissue samples.

Week 9: Align ground truth AAV measurement data (from Week 8) with the corresponding acoustic emission data logs.

Week 10: Begin building and training the baseline machine learning model using the newly structured dataset (mapping acoustic features to AAV delivery).

Week 11: Test and evaluate model accuracy; perform iterative adjustments to the selected features or model parameters.

Week 12: Finalize data analysis; synthesize findings into visualizations (graphs, performance metrics) for the final report.

Week 13: Draft, review, and finalize the summative written report for submission by the end of Reading Period.